

HAROUN BEN AMEUR

Software Engineer

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Profile

Software Engineer passionate about **web development** and **artificial intelligence**. Experienced in designing full-stack web applications, integrating complex systems, and building AI-based solutions. I design solutions integrating **LLMs** and **AI models**, **multi-component architectures**, **RAG**, **text generation**, **real-time systems**, and **intelligent automation** to build efficient, reliable, and innovative applications.

Experience

Final Year Internship – Talan Tunisia February 2026 – June 2026

Position: Fullstack & AI Developer
Project: SmartInvest – Intelligent Financial Investment Platform based on Multi-Agent AI
Technologies: Spring Boot, FastAPI, React, TypeScript, PostgreSQL, Redis, Kafka, WebSocket, Docker, Azure VM, LangGraph, LangChain, Chronos, FinBERT, LLM, Prometheus, Grafana, Langfuse
Description: Designed and developed an intelligent financial investment platform enabling market data analysis, portfolio tracking, financial news sentiment analysis, market trend prediction, and automatic analytical report generation. Implemented an architecture composed of a React frontend, a secure Spring Boot backend, and a FastAPI AI service orchestrating a multi-agent pipeline. Integrated Kafka, Redis, and WebSocket for real-time processing, as well as Prometheus, Grafana, and Langfuse for monitoring and observability. Containerized the application with Docker and set up a CI/CD pipeline with Jenkins for automated deployment on a Microsoft Azure virtual machine.

Internship – Proxym-IT July 2025

Position: AI Developer Intern
Project: FSD File Analysis Platform with Automatic Test Case Generation
Technologies: Python, Django, Elasticsearch, Kibana, Angular, Google Gemini, RAG, AI (Generative Models)

- Developed a platform to automate test case generation from Functional Specification Documents (FSD).
- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline combining document retrieval and generative AI for accurate test case generation.
- Designed a Django backend to manage file processing and semantic validation.

Internship – Proxym-IT August 2024

Position: Backend Developer – Data Engineering Intern
Project: Analysis and Reporting on Project Progress via Git and Elasticsearch
Technologies: Git, GitHub, Elasticsearch, Kibana, Python, Cohere

- Integrated GitHub with Elasticsearch to enhance data management and analysis.
- Implemented fuzzy search and semantic analysis for improved project tracking and visualization.
- Generated performance insights through dynamic dashboards in Kibana.

Projects

Issatso - Classroom Reservation Management System for Department Heads
Technologies: **React, Bootstrap, Redux, Spring Boot, PostgreSQL, WebSockets**
Project: Development of a web application enabling department heads to manage classroom reservations for professors according to specific time slots. The system also allowed professors to submit reservation requests and receive real-time notifications through WebSockets, ensuring efficient communication and scheduling within academic departments.

SmartPath – Intelligent Learning Platform
Technologies: **Next.js, Tailwind CSS, Django REST Framework, PostgreSQL, WebSockets, Exa, LiveKit, Deepgram, Cerebras LLM**
Project: Design and development of an intelligent educational platform featuring AI chatbots and voice tutors enhanced with EVA and personas to improve response relevance and performance, integrating real-time speech recognition and text generation.

CarCare App – Automotive Appointment Scheduling
Technologies: **React, Redux, Django, PostgreSQL**
Project: Design and implementation of a scheduling platform for automotive dealerships, allowing clients to book services online with integrated time-slot management. The solution improved customer experience while optimizing workshop resource allocation.

PFE App - Classroom Distribution System
Technologies: **Python, Django, REST API, Genetic Algorithm, React, Redux, SQLite**
Project: Backend development for an intelligent scheduling system allocating classrooms for thesis defenses. A genetic algorithm optimized room distribution while considering teacher availability and constraints. Exposed via a REST API and integrated with a React frontend for efficiency.

HireSphere – Employment Platform
Technologies: **React.js, Node.js, Express.js, MongoDB**
Project: Design and development of a comprehensive recruitment platform enabling job posting management, candidate tracking, and the monitoring of the entire hiring process.

Brain Tumor Detection – Computer Vision Project
Technologies: **Python, TensorFlow, Keras, OpenCV, NumPy, Matplotlib, MobileNetV2**
Project: Development of a convolutional neural network (CNN) model for automatic brain tumor detection from MRI images. Implemented image preprocessing and data augmentation using **ImageDataGenerator**, and compared performance between a custom CNN and a transfer learning model (MobileNetV2) to improve classification accuracy.

Education

Software Engineering Program – ISSAT Sousse September 2021 – July 2026
• Focused on full-stack development, AI integration, and project-based learning.

Baccalaureate in Technical Sciences (High Honors) – Mahmoud Mesaadi High School, Nabeul July 2021
• Achieved top grades with excellence in mathematics and physics.

Technical Skills

- **Programming Languages:** JavaScript, TypeScript, Python, Java, SQL
- **Frontend:** React, Angular, Next.js, Bootstrap, Tailwind CSS
- **State & Data Management:** Redux, Zustand, TanStack Query
- **Backend:** Spring Boot, Django, Django REST Framework, FastAPI, Express.js
- **Databases:** PostgreSQL, MySQL, MongoDB
- **AI & Data:** LangChain, LangGraph, RAG, LLM, FinBERT, Chronos, TensorFlow, Keras, OpenCV, Pandas, NumPy
- **Real-Time & Messaging:** WebSocket, Kafka
- **DevOps & Cloud:** Docker, Docker Compose, Azure VM, Jenkins
- **Monitoring & Observability:** Prometheus, Grafana, Langfuse, Spring Boot Actuator, Kibana
- **Search & Indexing:** Elasticsearch
- **Dev Tools:** Git, GitHub, Postman

Certifications

Building LLM Applications with Prompt Engineering – NVIDIA (2025)
• Hands-on training on large language models, prompt design, and application development using generative AI.

Deep Learning Certificate – NVIDIA (2025)
• Training on neural networks, model optimization, and AI workflows.

Advanced Fullstack Web Development (MERN) – Orange Digital Center (2025)
• Comprehensive certification in fullstack development with MongoDB, Express.js, React, and Node.js.

Associative Experience

Vice President – NATEG ISSATSO (North American Tunisian Engineers Group)
• Organized events and hackathons; enhanced public speaking and leadership skills.

Member – ONET (National Organization for Children in Tunisia)
• Participated in community projects, fostering teamwork and adaptability.

Languages

- Arabic: Fluent
- French: Intermediate
- English: Intermediate
- Spanish: Beginner